

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

**Annexure A
Scenario-Based**

Code of Conduct:

I P. Harsha vardhan (192411179) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :

ANSWERS TO SCENARIO-BASED QUESTIONS

Question 1: Smart Campus – Subnetting 192.168.100.0/24

Understanding of the Scenario:

A newly established Smart Campus has been allocated the network block 192.168.100.0/24, which provides 256 total addresses ($256 = 2^8$, since 8 host bits are available in a /24 network). This single block must be divided into three smaller subnets of different sizes to serve the School of Computing, School of Electronics, and School of Mechanical Engineering, based on their expected number of devices. Since /25, /26, and /27 masks are already fixed for each department, the task is to carve out contiguous, non-overlapping address ranges from the parent block in decreasing order of size, so that the larger department is placed first and each subsequent department starts immediately after the previous one ends.

Key Issues Identified:

- (1) The subnets must not overlap and must fit within the 192.168.100.0/24 boundary.
- (2) Larger blocks should be allocated first to avoid fragmentation.
- (3) Each subnet mask must be converted to understand how many host bits remain, which decides the block size and the usable host count.
- (4) Any space left after allocation must be tracked as free/unused for future growth.

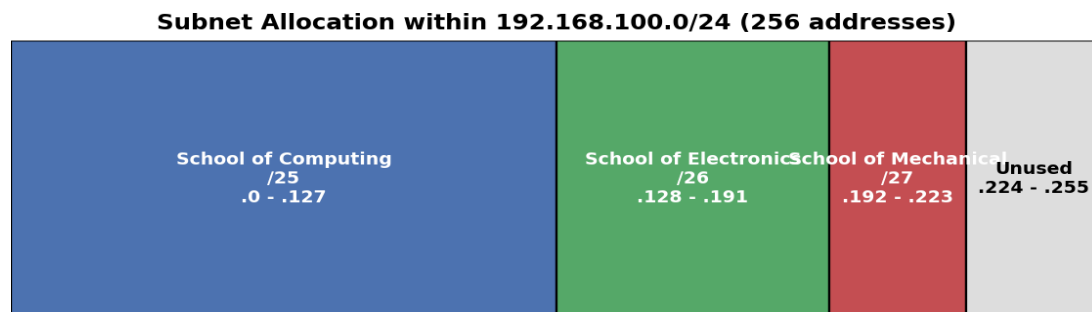
Subnet Mask Breakdown (Binary View):

/25 → 11111111.11111111.11111111.10000000 → Block size = $2^{(32-25)} = 2^7 = 128$ addresses

/26 → 11111111.11111111.11111111.11000000 → Block size = $2^{(32-26)} = 2^6 = 64$ addresses

/27 → 11111111.11111111.11111111.11100000 → Block size = $2^{(32-27)} = 2^5$
= 32 addresses

Allocation Diagram:



1. Network Address and Broadcast Address (Application of Concepts):

School of Computing (192.168.100.0/25): starts at .0, occupies 128 addresses →
Network Address = 192.168.100.0, Broadcast Address = 192.168.100.127.

School of Electronics (192.168.100.128/26): starts immediately after Computing ends, occupies 64 addresses → Network Address = 192.168.100.128, Broadcast Address = 192.168.100.191.

School of Mechanical Engineering (192.168.100.192/27): starts after Electronics ends, occupies 32 addresses → Network Address = 192.168.100.192, Broadcast Address = 192.168.100.223.

2. Valid Host IP Address Range:

School of Computing: 192.168.100.1 – 192.168.100.126 (excluding network .0 and broadcast .127).

School of Electronics: 192.168.100.129 – 192.168.100.190 (excluding network .128 and broadcast .191).

School of Mechanical Engineering: 192.168.100.193 – 192.168.100.222 (excluding network .192 and broadcast .223).

3. Number of Usable Host Addresses:

School of Computing (/25): $2^7 - 2 = 126$ usable hosts.

School of Electronics (/26): $2^6 - 2 = 62$ usable hosts.

School of Mechanical Engineering (/27): $2^5 - 2 = 30$ usable hosts.

(Two addresses are always subtracted per subnet — one for the network address and one for the broadcast address — because neither can be assigned to a host.)

4. Remaining Unused IP Address Range (Decision Making):

Total addresses consumed by the three departments = $128 + 64 + 32 = 224$ out of 256. The remaining unused block is 192.168.100.224 – 192.168.100.255, which is exactly a /27-sized block of 32 addresses. This is reserved for future departments or expansion, following good design practice of not exhausting the entire allocated block immediately.

Question 2: Multinational Software Company – Subnetting 10.10.0.0/16

Understanding of the Scenario:

The company has been assigned the private block 10.10.0.0/16, giving 65,536 total addresses (2^{16}). Four departments — Software Development, Quality Assurance, Human Resources, and Executive Management — must be allocated subnets of sizes /24, /25, /26 and /27 respectively, again from largest to smallest, so that each subsequent subnet begins right where the previous one ends.

Key Issues Identified:

(1) Even though the parent block is very large (/16), subnets must still be allocated contiguously and without overlap. (2) The largest subnet (/24) must be placed first. (3) Because the block is large, most of the /16 space remains unused after only four small subnets are carved out — this must be clearly reported as available address space for future scaling.

Subnet Mask Breakdown (Binary View):

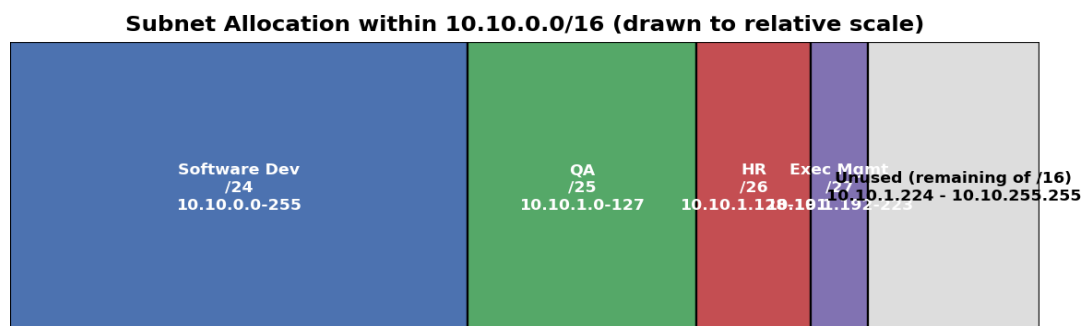
/24 → block size = $2^{(32-24)} = 2^8 = 256$ addresses

/25 → block size = $2^{(32-25)} = 2^7 = 128$ addresses

/26 → block size = $2^{(32-26)} = 2^6 = 64$ addresses

/27 → block size = $2^{(32-27)} = 2^5 = 32$ addresses

Allocation Diagram:



1. Network Address and Broadcast Address:

Software Development (10.10.0.0/24): Network Address = 10.10.0.0, Broadcast Address = 10.10.0.255.

Quality Assurance (10.10.1.0/25): Network Address = 10.10.1.0, Broadcast Address = 10.10.1.127.

Human Resources (10.10.1.128/26): Network Address = 10.10.1.128, Broadcast Address = 10.10.1.191.

Executive Management (10.10.1.192/27): Network Address = 10.10.1.192, Broadcast Address = 10.10.1.223.

2. Valid Host IP Address Range:

Software Development: 10.10.0.1 – 10.10.0.254.

Quality Assurance: 10.10.1.1 – 10.10.1.126.

Human Resources: 10.10.1.129 – 10.10.1.190.

Executive Management: 10.10.1.193 – 10.10.1.222.

3. Number of Usable Hosts:

Software Development (/24): $2^8 - 2 = 254$ usable hosts.

Quality Assurance (/25): $2^7 - 2 = 126$ usable hosts.

Human Resources (/26): $2^6 - 2 = 62$ usable hosts.

Executive Management (/27): $2^5 - 2 = 30$ usable hosts.

4. Unused IP Address Range Remaining (Decision Making):

The four departments together consume $256 + 128 + 64 + 32 = 480$ addresses out of the 65,536 available in the /16 block. The remaining unused range is 10.10.1.224 – 10.10.255.255, i.e. the vast majority of the /16 space, which is kept available for

future departments, additional offices, or network growth without needing to re-plan the existing allocation.

Question 3: Smart Hospital – IPv4 to IPv6 Migration

Understanding of the Scenario:

The hospital is migrating to IPv6 to support a very large and growing number of connected medical devices and IoT sensors, since IPv4's limited address space (about 4.3 billion addresses) cannot scale to meet modern device density. The hospital has been assigned the /64 network 2001:0db8:abcd:0000:0000:0000:0000:0000, which is the standard subnet size recommended for IPv6 LANs.

Key Issues Identified:

- (1) Correctly compressing a long IPv6 address using the two official compression rules.
- (2) Being able to reverse the process and expand a compressed address back to its full 128-bit hexadecimal form.
- (3) Understanding that in a /64 prefix, exactly the first 64 bits identify the network and the last 64 bits identify the interface (host).
- (4) Calculating the enormous address space a /64 subnet provides.

1. Compressed Notation:

Applying IPv6 compression rules — (a) leading zeros within each 16-bit group can be omitted, and (b) one single contiguous run of all-zero groups can be replaced with '::' — the address compresses to: 2001:db8:abcd::/64

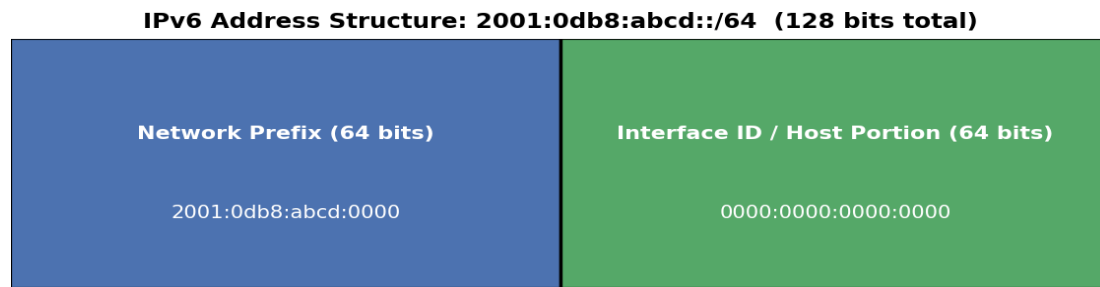
2. Expanded (Full Hexadecimal) Representation:

Reversing the compression, each group is restored to its full 4-hex-digit form: 2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix using the /64 Prefix Length:

A /64 prefix means the first 64 bits (the first four groups) form the network portion, and the remaining 64 bits (last four groups) form the interface identifier. Network Prefix = 2001:0db8:abcd:0000 (equivalently 2001:db8:abcd::/64).

IPv6 Address Structure Diagram:



4. Total Number of Host Addresses Supported within the Subnet:

Host bits available = 128 total bits – 64 network bits = 64 bits. Total addresses = $2^{64} = 18,446,744,073,709,551,616$ addresses.

5. Total Number of Possible Host Addresses Available within this Network (Decision Making):

Unlike IPv4, IPv6 does not reserve a separate broadcast address, and the all-zero/all-one host addresses within a subnet are not disallowed in the same restrictive way — so effectively the entire 2^{64} (18,446,744,073,709,551,616) address space within the /64 subnet is available for assigning to medical devices, IoT sensors, and other hospital equipment. This provides more than enough headroom for the hospital to expand its connected-device footprint indefinitely without ever needing to re-subnet.

Question 4: ISP Core Router – Routing Table Lookup

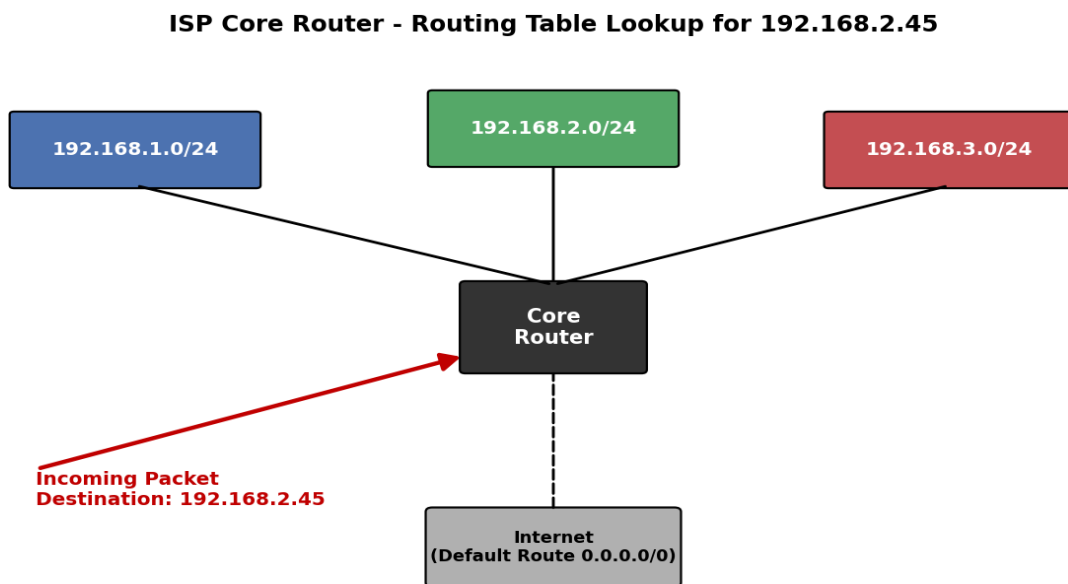
Understanding of the Scenario:

An ISP's core router connects multiple customer networks and maintains a routing table with three directly connected subnets: 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24. When a packet destined for 192.168.2.45 arrives, the router must consult its routing table and decide which interface to forward the packet out of, using the longest prefix match algorithm.

Key Issues Identified:

(1) Correctly identifying which /24 subnet the destination address 192.168.2.45 belongs to. (2) Applying longest-prefix-match logic when multiple entries could partially resemble the destination. (3) Understanding what happens if no route matches — the role of the default route.

Network Topology and Packet Flow:



1. Destination Subnet:

192.168.2.45 lies within the range 192.168.2.0 – 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry (Application of Concepts):

Comparing the destination address against all three table entries using longest prefix match, only 192.168.2.0/24 matches, since the first 24 bits of 192.168.2.45 (192.168.2) exactly match this entry's network portion.

3. Next-Hop Network/Interface (Decision Making):

Because 192.168.2.0/24 is a directly connected subnet on the router, the packet is forwarded out of the router interface that is physically/logically configured with an IP address belonging to 192.168.2.0/24 (i.e., delivered directly to the destination network rather than to another next-hop router).

4. Default Route (if no match is found):

If the destination address did not match any of the three specific entries, the router would forward the packet using its default route, written as 0.0.0.0/0, which typically points toward the ISP's upstream gateway or the wider internet, ensuring that traffic to unknown networks is still forwarded onward rather than dropped.

Question 5: Two Office Departments via Layer-3 Router

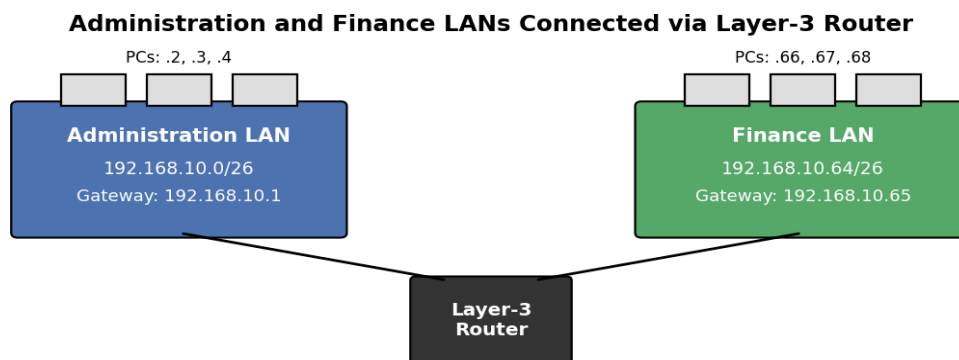
Understanding of the Scenario:

A company has two departments — Administration and Finance — connected through a Layer-3 router, which allows inter-department (inter-VLAN/inter-subnet) communication. Each department has been given a /26 subnet from the 192.168.10.0 block: Administration LAN = 192.168.10.0/26 and Finance LAN = 192.168.10.64/26. The router interfaces act as the default gateway for each LAN.

Key Issues Identified:

(1) Calculating the network and broadcast address for each /26 subnet. (2) Identifying valid host ranges so IP addresses can be assigned without conflict. (3) Confirming that the given gateway addresses (192.168.10.1 and 192.168.10.65) are valid, in-range host addresses for their respective subnets. (4) Making sure suggested computer IP addresses fall inside the valid host range and do not clash with the gateway address.

Network Topology Diagram:



1. Network Address and Broadcast Address:

Administration LAN (192.168.10.0/26): Network Address = 192.168.10.0, Broadcast Address = 192.168.10.63.

Finance LAN (192.168.10.64/26): Network Address = 192.168.10.64, Broadcast Address = 192.168.10.127.

2. Valid Host IP Address Range:

Administration LAN: 192.168.10.1 – 192.168.10.62 (62 usable hosts, since $2^6 - 2 = 62$).

Finance LAN: 192.168.10.65 – 192.168.10.126 (62 usable hosts).

3. Suggested IP Addresses for Three Computers in Each LAN (Application of Concepts):

Administration LAN: PC1 = 192.168.10.2, PC2 = 192.168.10.3, PC3 = 192.168.10.4 (all within the valid range and distinct from the gateway address .1).

Finance LAN: PC1 = 192.168.10.66, PC2 = 192.168.10.67, PC3 = 192.168.10.68 (all within the valid range and distinct from the gateway address .65).

4. Default Gateway Each LAN Should Use (Decision Making):

Hosts in the Administration LAN should be configured with default gateway 192.168.10.1, which is the router's interface IP address for that subnet.

Hosts in the Finance LAN should be configured with default gateway 192.168.10.65, which is the router's interface IP address for that subnet.

Both gateway addresses are correctly chosen — they fall inside their respective valid host ranges and are reserved from being assigned to end-user computers, so all inter-department and internet-bound traffic is properly routed through the Layer-3 router.

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand